

American University of Central Asia

Software Engineering Department

COM-_____, _____

1. Course Information Course Codes COM - 424.1

Course ID 3953

Prerequisites for the Course "Information Security"

Basic Knowledge That Can Be Useful:

- **Computer Science Fundamentals:** Understanding the functioning of computers, networks, operating systems, and software.
- **Programming:** Familiarity with at least one programming language (e.g., Python, C++) to comprehend the principles of software operation and vulnerabilities.
- **Mathematics:** Knowledge of basic probability theory, mathematical logic, and discrete mathematics is valuable for risk analysis and cryptographic algorithms.
- **Networking:** Understanding the principles of computer networks, data transmission protocols (TCP/IP), and network security.
- **Operating Systems:** Familiarity with the architecture and functionality of different operating systems (Windows, Linux, macOS).
- **Databases:** Knowledge of database principles and SQL for securing stored information.

Skills That Can Be Beneficial:

- **Analysis:** Ability to analyze information, identify patterns, and make data-driven decisions.
- **Problem-Solving:** Capability to identify and resolve issues in systems and software.
- **Logical Thinking:** Ability to construct logical sequences and draw conclusions.
- **Critical Thinking:** Skill in evaluating information, assessing its reliability, and making well-founded decisions.

Credits 6

Time and Place Lecture: _____

Course

Materials:

https://drive.google.com/drive/u/2/folders/14_kGbJW1h5Fb0I9QuxyrgfB7siuKGSH7

Contact Information: **Ph.D, Associate Professor** Gulzada Esenalieva, esenalieva_gu@auca.kg

Syllabus: Introduction to Information Security

Course Information

- **Course Code:** 424.1
 - **Credits:** __6__
 - **Prerequisites:** None
 - **Course Format:** Lectures and laboratory work
 - **Time and Location:** To be announced
 - **Course Materials:** Provided through the Learning Management System (LMS) and additional online resources
https://drive.google.com/drive/u/2/folders/14_kGbJW1h5Fb0I9QuxyrgfB7siuKGSH7
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Contact Information

- **Instructor:** Ph.D, Associate Professor Gulzada Esenalieva
- **Email:** esenalieva_gu@auca.kg
- **Office Hours:** By appointment (offline: Monday Tuesday Wednesday; online via Telegram @HDGulzada)
- **Teaching Assistant:** _____
- **Email:** _____

Course Overview

This course provides foundational knowledge in Information Security. Topics include core concepts such as terminal, confidentiality, integrity, and availability, risk management, types of cyber threats, basics of cryptography, network security, and secure programming practices. Students will gain practical skills in identifying vulnerabilities, implementing basic security measures, and understanding the ethical and legal aspects of cybersecurity.

Course Topics

- **Week 1-2:** Terminal usage
 - **Week 3-4:** Introduction to Information Security and Overview of Cyber Threats
 - **Week 5-6:** Confidentiality, Integrity, and Availability (CIA Triad)
 - **Week 7-8:** Basics of Network Security, Fundamentals of Cryptography
 - **Week 9-10:** Secure Programming Practices (CISCO)
 - **Week 11-12:** Risk Assessment and Management
 - **Week 13-14:** Ethical and Legal Aspects of Information Security
 - **Week 15:** Final Project and Course Review
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Learning Outcomes

By the end of the course, students will be able to:

1. Understand terminal shell and main instructions

2. Understand the fundamental principles of information security, including the CIA triad (Confidentiality, Integrity, Availability).
 3. Identify and classify various types of cyber threats and vulnerabilities.
 4. Apply basic cryptographic techniques to secure information.
 5. Evaluate and implement security measures to protect networks and systems.
 6. Develop secure programming practices to mitigate risks in applications.
 7. Conduct risk assessments and propose strategies for effective risk management.
 8. Analyze ethical and legal aspects in information security scenarios.
 9. Effectively explain security concepts to diverse audiences.
 10. Work collaboratively in teams, demonstrating cohesion and effective interaction in group projects.
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Assignments and Exams

Laboratory Work and Projects

- **Weekly Laboratory Sessions:** Hands-on exercises using online resources to build practical experience.
- **Final Project:** Analysis of a real-world security scenario and proposal of effective solutions.

Laboratory Topics:

1. **Comprehending UNIX commands:**
 - Overview of main UNIX terminal commands.
 - Creating custom UNIX commands.
2. **User management in Linux:**
 - Overview of user management in Linux terminal.
 - Creating new user and groups.
3. **Permissions in Linux:**
 - Overview of permissions in Linux.
 - Changing file permissions.
4. **Nginx Configuration:**
 - Description of Nginx.
 - Creating configuration file in Nginx.
5. **Vim editor:**
 - Overview of Vim editor.
 - Importance of vim editor.
6. **Crontab in Linux:**
 - Description of Crontab utility.
 - Creating crontab tasks.
7. **Package management:**
 - Package manager.
 - Main commands.

8. **Installing packages from source:**
 - Reasons to install from source.
 - Types of installation.
9. **Introduction to Cyber Threats:**
 - Overview of common internet threats.
 - Basic protection methods, including antivirus software and firewall configuration.
10. **Cryptography Basics:**
 - Practical implementation of the Caesar cipher to understand encryption and decryption principles.
11. **Network Connections:**
 - Configuring basic network parameters on a personal computer.
 - Examining Wi-Fi network security and vulnerabilities.
12. **Password Management:**
 - Creating and managing secure passwords.
 - Using password managers and enabling two-factor authentication.
13. **Phishing Attack Detection:**
 - Analyzing phishing email examples.
 - Developing practical recommendations for identifying and preventing phishing attempts.
14. **Keylogger:**
 - Overview of key logging.
 - Developing a script that logs keys using Python.

Advanced Laboratory Work:

- **Unix Utilities Analysis:**
 - Students will analyze standard Unix utilities and attempt to implement them independently.
 - Implementation will involve direct system calls to closely interact with the operating system kernel.
 - These tasks will help students explore the interfaces between the OS kernel and user applications.

Completed Lab projects have to be uploaded in either Github or Gitlab

Projects:

Students are required to complete four projects during the course:

1. **Toy Shell Development:** Creating a simple command-line interface.
2. **Linux Kernel Modification:** Exploring and customizing basic kernel functionality.
3. **File System Driver Utilities:** Developing tools for basic file system operations.
4. **Web Server Optimization:** Enhancing a web server using low-level OS APIs.

These assignments and projects aim to reinforce theoretical knowledge through practical application and foster a deeper understanding of core concepts in information security.

Exams

- **Midterm Exam (_____):** Covers topics from Weeks 1 to 7, including foundational concepts such as the CIA triad, cryptography basics, network security, and early lab assignments.
 - **Final Exam (_____):** Focuses on the final project, including its presentation and defense. Students must demonstrate their ability to apply learned concepts to analyze security scenarios, propose effective solutions, and articulate their findings clearly.
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Assessment Criteria

Component	Weight (%)
Laboratory Work	30
Final Project	20
Midterm Exam	25
Final Exam	25

Grading scale:

- [94%–100] %: A
 - [90%–94) %: A-
 - [87%–90) %: B+
 - [83%–87) %: B –
 - [80%–83) %: B-
 - [77%–80) %: C+
 - [73%–77) %: C –
 - [70%–73) %: C-
 - [67%–70) %: D+
 - [63%–67) %: D
 - [60%–63) %: D-
 - Less than 60%: F
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Course Materials

Topics for final

<https://docs.google.com/document/d/1cGUkFKaxVDsjgh9YeU5-b-tgZ3gNHti0/edit>

<https://docs.google.com/document/d/1mgIrCwzt-hevxKgBr-5jgQ3S4Fml2BQB/edit>

- <https://www.netacad.com/courses/ethical-hacker?courseLang=en-US>
 - Recommended Book: Principles of Information Systems Hardcover -14th Edition, CEHTM v12 Certified Ethical Hacker Study Guide
 - Additional materials and resources will be available at _____.
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Course Guidelines and What to Expect

- **Show up!** Active participation is key. Missing more than 8 classes without a good reason might mean you have to leave the course. If you're working in your field, we can talk about a flexible schedule, but you'll still need to finish the final project and bring a note from your job.
- **Don't be late:** Get your work in on time. If something really important comes up, let us know ahead of time.
- **Play fair:** No cheating or copying. Using AI to write your code isn't cool. We'll have to take action if you do.
- **Speak up:** Share your thoughts in class and lab. Your input matters!

Schedule and Communication

- Weekly updates, assignments, and announcements will be posted in _____.
- Students are encouraged to use the _____ forums for discussions and questions.

Final Project Details

- Students work in teams to identify vulnerabilities in a hypothetical organization's IT infrastructure and propose security measures.
- Deliverables: Project, presentation, and CISCO certification.

For all questions, please contact the teacher or teaching assistant by email, esenalieva_gu@auca.kg, tg:@HDGulzada

Rules

Students are required to follow the rules of conduct of the Software Engineering Department and the American University of Central Asia.

- Participation Active work during the class may be awarded with extra points at the instructor's discretion. Poor student performance during a class can lead to points being deducted from the final grade. Instructors may conduct pop-checks during classes at random without prior notice. Students MUST be ready for every class in order not to lose points. Students absent without a good reason from such classes with graded work will also lose points unless it is force-majeure circumstances. Instructors must be notified in advance about why a student is absent not to lose points.

- Questions We believe that a question from one student is most likely a question that other students are also interested in. That is why we encourage students to use the online discussion board of the LMS (Learning Management System) that you use (e.g., AUCA e-Course System) to ask questions in public that other students can see and answer. Do not post the complete source code for any task on the LMS discussion board. You will get zero for that work for any such public post. Do not ask generic questions about your code to know why it does not work. Please spend some time thinking about your code, debugging it.

- Late Policy Late submissions and late exams are not allowed. Exceptions may be made at the professor's discretion only in force-majeure circumstances. If you got ill, got severe personal issues, got problems with your computer or the Internet, you MUST notify instructors at least 24 hours in advance. Otherwise, we will not give you an extension. We will consider that you were procrastinating until the very last day. We will also not be giving more than one emergency extension throughout the course. Six hours before the deadline for any work on the course,

instructors will go into a silent mode. No questions will be answered about the work that has to be submitted, no requests to have office hours will be considered. However, at any other work time before the deadline, we will try our best to answer your questions and help you through Zoom or in our office.

- **Exam and Task Submission Ceremonies** Students **MUST** follow exam and task submission ceremonies. It means they **MUST** strictly follow all the rules specified by the instructors in written or verbal form. Failure to do so will result in lost points. Throughout your career, you will have to work with various supporting documents (contracts, timesheets, etc.). It is a good idea to start learning to work with such documents accurately early. We will remove points for not following these rules or even refuse to accept your exam defense or tasks submitted to us. We will also give zero for not following deadlines or the strict exam timing rules.

- **Administrative Drop** Instructors have a right to drop a student from the course for non-attendance. If you have five classes or more missed without an excuse, the faculty may consider dropping you by giving you the X grade. 11.6 Incomplete Grade Similar to the policy for late exams, the grade I may be awarded only in highly exceptional circumstances. Students **MUST** initiate a discussion about receiving an I grade with the instructors well in advance and **NOT** during the last week before final exams.

- **Academic Honesty Plagiarism** is the act of copying or stealing someone else's words or ideas and presenting them as one's own. This definition encompasses various task elements, including but not limited to program code, comments, software documentation, abstracts, reports, diagrams, and statistical tables.

- The following are examples of plagiarism in the context of a Software Engineering course:

1. Presenting code written by others as your own
2. Purchasing code, software, or any project-related content from online platforms or other sources and submitting it as your own creation.
3. Using algorithms, patterns, or architectural designs without acknowledging the source
4. Incorporating code snippets, sentences, design patterns, or any intellectual content from sources, published or unpublished, without proper citation
5. Modifying someone else's code or design (e.g., changing variable names, changing the structure of the code) and claiming it as original
6. Utilizing graphics, data sets, audio, video, or other elements from external works without proper acknowledgment. Engaging in plagiarism is not only unethical but also undermines the educational process. The consequences for plagiarism in this course for all parties involved are as follows:

First instance: The students will receive a grade of zero for the plagiarized work, and a report will be filed with the Registrar's Office.

- Second instance: The students will receive an F grade for the entire course.

It's important to note that both parties involved in plagiarism—the one who plagiarizes and the one whose work was copied—will face equal consequences. This underscores the imperative for honest students to exercise caution in ensuring the security of their work. It is the student's

responsibility to guarantee that their assignments, code, or any related content can only be accessed by them and the course instructors. Sharing, unintentionally exposing, or not securely storing one's work can lead to unintended consequences and sanctions. Students are advised against rote memorization of code for examinations. Relying solely on memorization is an ineffective learning strategy in programming. Examinations in this course may contain open-ended questions targeting the student's analytical and design skills, and memorization may lead to answers that are off-target and of subpar quality. In addition to the rules outlined in this syllabus, we abide by all global university policies concerning plagiarism. Should the global university rules evolve to be more consequential or stringent than what is stipulated here, those university-wide regulations will take precedence over our course-specific rules.

Legal simulation platforms designed for learning **ethical hacking, penetration testing (pentesting), and cybersecurity**. These platforms allow you to practice **finding vulnerabilities, hacking systems, and even conducting phishing attacks**, but **strictly within legal and educational purposes**.

Popular Platforms for Learning Hacking Techniques

- ♦ **Hack The Box** (hackthebox.com)

- One of the most popular platforms for hands-on hacking training. It offers labs for server exploitation, web application security, and phishing simulations.

- ♦ **TryHackMe** (tryhackme.com)

- Great for beginners, providing interactive learning paths on hacking fundamentals, social engineering, and phishing attacks.

- ♦ **RangeForce** (rangeforce.com)

- A cloud-based cybersecurity training platform that simulates real-world attacks, including phishing campaigns. Ideal for corporate security teams.

- ♦ **Hack This Site** (hackthissite.org)

- A free online platform offering challenges on various hacking techniques, including phishing simulations.

- ♦ **Social Engineer Toolkit (SET)** (trustedsec.com)

- A specialized tool for testing **social engineering tactics**, including phishing attacks and fake websites.

Important!

Ethical hacking is only legal for **educational and research purposes**. Using these skills **without permission** (e.g., real-world phishing attacks) **is illegal** and can result in criminal prosecution.

